

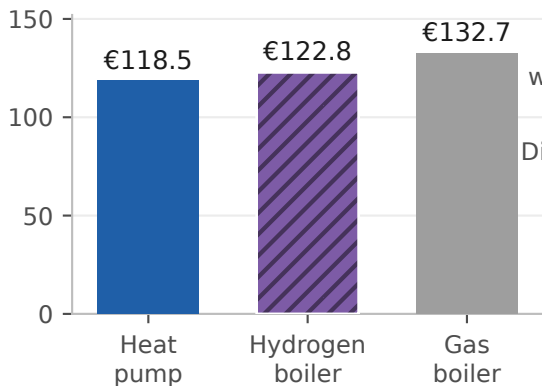
Heat pumps decarbonise Europe's homes, whereas hydrogen is a support-contingent capacity option

29 European markets (EU-27, UK, Switzerland)

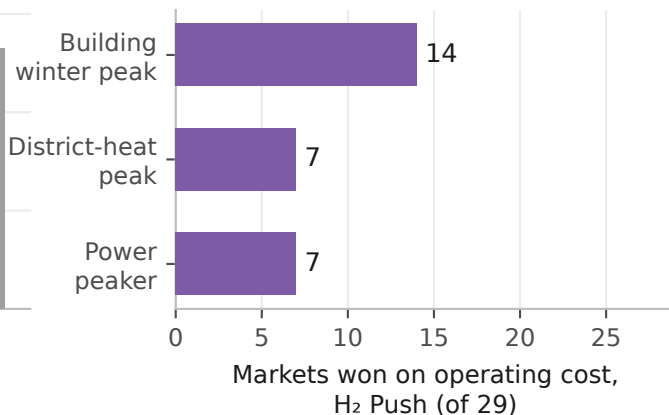
1,369-region NUTS3 demand surface · 2025 to 2050

2050 levelised cost of heat (€/MWh)

Base load: 2050 delivered cost



Peaking: dispatch wins



Heat pumps beat the hydrogen boiler in 22 of 29 countries, and 5 of the 7 exceptions are salt-cavern markets.

Counts charge both carriers symmetrically, including hydrogen's own last-mile network. No tested market reaches full peaker-capital recovery across the assumed scarcity margins. On the high-carbon path, hydrogen wins the power peak in seven salt-cavern markets, a different set from its base-load wins; the five that build require €31–63/kW-yr to be financeable.